ASEEMA S

QUANTUM COMPUTING • QUANTUM AI • RESEARCH

Avadi, Chennai, Tamil Nadu
Phone: 9789367963 | Email: aseema1652@gmail.com
LinkedIn: linkedin.com/in/aseema-s
GitHub: github.com/Aseema-s

PROFESSIONAL SUMMARY

Computer Science and Engineering student at Rajalakshmi Institute of Technology (GPA: 8.18 with a minor in Quantum Computing. Interested in quantum algorithms, quantum machine learning, and computational research. Experience includes Qiskit-based VQE molecular simulation, QAOA-oriented portfolio optimization, QML crop-yield prediction, and Python graph-algorithm applications.

EDUCATION

B.E. Computer Science and Engineering | Rajalakshmi Institute of Technology | 2024–2028 | GPA: 8.18

Minor in Quantum Computing | Rajalakshmi Institute of Technology

TECHNICAL SKILLS

Quantum Computing: Qiskit, PennyLane, QAOA, VQE, QUBO, quantum circuit design, quantum machine learning (QML)

Quantum Concepts: Quantum entanglement, density matrices, Bell's inequality, complementarity

Programming: Python, Java, C, C++, SQL

AI / ML: Machine learning, NLP, TensorFlow, PyTorch

Tools & Frameworks: Flask, Streamlit, Git, GitHub, VS Code, Linux

Computer Science: Data structures, object-oriented programming, discrete mathematics, graph algorithms

PROJECTS

- **Molecular Stability Simulation (VQE):** Simulated molecular stability using Qiskit and the Variational Quantum Eigensolver (VQE); authored research focused on energy optimization and drug-discovery applications.
- **Quantum Portfolio Optimization:** Built a quantum-finance portfolio optimization tool using Qiskit and Streamlit, exploring QAOA-based optimization.
- **Crop-Yield Prediction with QML:** Implemented quantum machine learning models for crop-yield prediction and agricultural forecasting.
- **AI Internship Recommender:** Developed a Flask-based matching prototype using natural language processing for the PM Internship Scheme.
- **Traffic Flow Optimizer:** Developed a Python tool using graph algorithms to analyze urban traffic flow and explore congestion reduction.

LEADERSHIP & RELEVANT ACTIVITIES

- **Quantum Computing Workshop (2025):** Hands-on training in QAOA, VQE, QUBO, and quantum machine learning.
- **World Quantum Day 2026 — Quinfosys Quantum Hackathon and Quantathon:** Participated in quantum-focused hackathon events.
- **Smart India Hackathon:** Collaborated with a six-member team to prototype solutions for national-level challenges.

CERTIFICATIONS & LEARNING

- **Data Science Foundation:** Great Learning
- **Machine Learning with PyTorch:** Training

LANGUAGES

English (Fluent) • Tamil (Fluent) • Arabic (Intermediate) • Japanese (Basic) • Hindi (Basic) • Urdu (Intermediate)